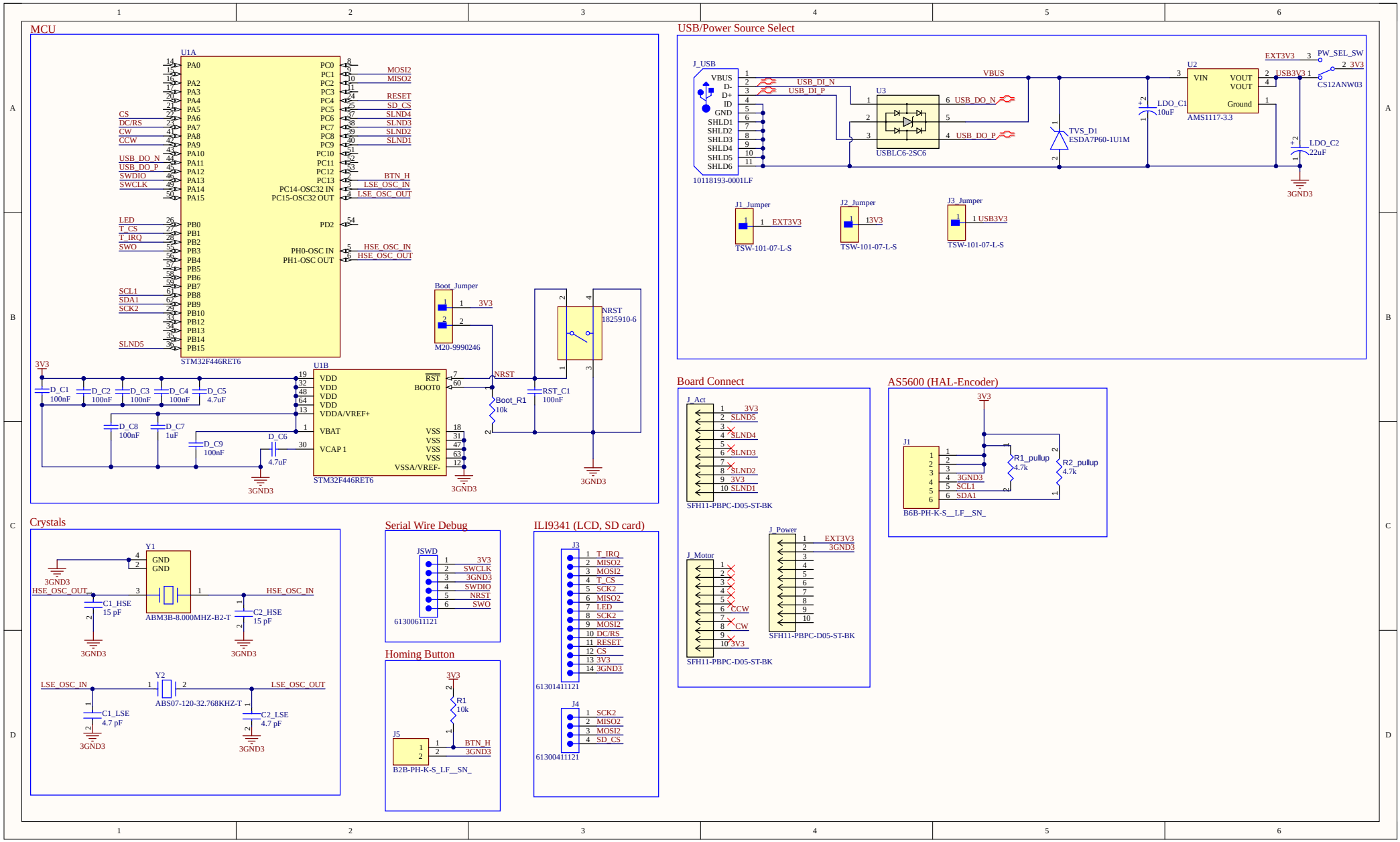
MCU
U1A
STM32F446RET6
U1B
STM32F446RET6
Boot_Jumper
M20-9990246
NRST
1825910-6
Boot_R1
10k
RST_C1
100nF
3V3
3GND3
USB/Power Source Select
J_USB
10118193-0001LF
USB_DI_N
USB_DI_P
U3
USBLC6-2SC6
USB_DO_N
USB_DO_P
VBUS
TVS_D1
ESDA7P60-1U1M
LDO_C1
10uF
U2
AMS1117-3.3
EXT3V3
PW_SEL_SW
USB3V3
CS12ANW03
LDO_C2
22uF
J1_Jumper
TSW-101-07-L-S
J2_Jumper
J3_Jumper
Board Connect
J_Act
SFH11-PBPC-D05-ST-BK
J_Motor
J_Power
AS5600 (HAL-Encoder)
J1
B6B-PH-K-S__LF__SN_
R1_pullup
4.7k
R2_pullup
4.7k
Crystals
Y1
ABM3B-8.000MHZ-B2-T
HSE_OSC_IN
HSE_OSC_OUT
C1_HSE
15 pF
C2_HSE
15 pF
Y2
ABS07-120-32.768KHZ-T
LSE_OSC_IN
LSE_OSC_OUT
C1_LSE
4.7 pF
C2_LSE
4.7 pF
Serial Wire Debug
JSWD
61300611121
Homing Button
R1
10k
J5
B2B-PH-K-S_LF__SN_
ILI9341 (LCD, SD card)
J3
61301411121
J4
61300411121

Comment	Description	Designator	Footprint	LibRef	Quantity
M20-9990246	CONN HEADER VERT 2POS 2.54MM	Boot_Jumper	FP-M20-9990246-MFG	CMP-15831-000009-1	1
Resistor 10k +/-1% 0603 100 mW	Chip Resistor, 10 KOhm, +/- 1%, 01 W, -55 to 155 degC, 0603 (1608 Metric)	Boot_R1, R1, R1_pullup, R2_pullup	RESC1608X55X25ML10T15	CMP-009-00017-5	4
15 pF	Multi-Layer Ceramic Capacitor 15pF C0G ±2% 0402 Paper T/R	C1_HSE, C2_HSE	FP-GJM155-0_05-MFG	CMP-2008-03803-2	2
4.7 pF	Multi-Layer Ceramic Capacitor 15pF C0G ±2% 0402 Paper T/R	C1_LSE, C2_LSE	FP-GJM155-0_05-MFG	CMP-2008-03803-2	2
100nF		D_C1, D_C2, D_C3, D_C4, D_C8, D_C9, RST_C1	CAPC1608X90X35NL15T15	CMP-2006-02299-1	7
4.7uF		D_C5, D_C6	CAPC1608X90X35NL15T15	CMP-2006-02299-1	2
1uF		D_C7	CAPC1608X90X35NL15T15	CMP-2006-02299-1	1
B6B-PH-K-S_LF_SN_	Connector	J1	SHDR6W50P0X200_1X6_1390X450X600P	B6B-PH-K-S_LF_SN_	1
TSW-101-07-L-S	CONN HEADER VERT 1POS	J1_Jumper, J2_Jumper, J3_Jumper	FP-TSW-101-07-L-S-MFG	CMP-02766-001775-1	3
61301411121	THT Vertical Pin Header WR-PHD, Pitch 2.54 mm, Single Row, 14 pins	J3	61301411121	CMP-1502-01091-4	1
61300411121	THT Vertical Pin Header WR-PHD, Pitch 2.54 mm, Single Row, 4 pins	J4	61300411121	CMP-1502-01064-4	1
B2B-PH-K-S_LF_SN_	Connector	J5	SHDR2W50P0X200_1X2_590X450X600P	B2B-PH-K-S_LF_SN_	1
SFH11-PBPC-D05-ST-BK	Description	J_Act, J_Motor, J_Power	CONN_SFH11-xxxC-D05-ST-XX_SUL	SFH11-PBPC-D05-ST-BK	3
10118193-0001LF	Micro USB B Type Receptacle, -55 to 85 degC, 5-Pin SMD, RoHS, Tape and Reel	J_USB	FCI-10118193-0001LF5_V	CMP-2000-05119-1	1
61300611121	THT Vertical Pin Header WR-PHD, Pitch 2.54 mm, Single Row, 6 pins	JSWD	61300611121	CMP-1502-01070-4	1
10uF	10uF 10V ±20% 3Ω 1206 SMD Tantalum Capacitor	LDO_C1	FP-TAJA-MFG	CMP-2000-07219-2	1
22uF	10uF 10V ±20% 3Ω 1206 SMD Tantalum Capacitor	LDO_C2	FP-TAJA-MFG	CMP-2000-07219-2	1
1825910-6	Tact Switch, SPST-NO, 0.05 A, -35 to 85 degC, 4-Pin THD, RoHS, Bulk	NRST	TECO-1825910-6_V	CMP-1684-00021-1	1
CS12ANW03	SWITCH SLIDE SPDT 3A 125V	PW_SEL_SW	FP-CS12ANW03-MFG	CMP-31746-000002-1	1
ESDA7P60-1U1M	TVS DIODE 5V 11.6V 1610	TVS_D1	FP-1610-MFG	CMP-12225-000013-1	1
STM32F446RET6	IC MCU 32BIT 512KB FLASH 64LQFP	U1	FP-LQFP64-5W-MFG	CMP-12105-000007-1	1
AMS1117-3.3	LDO Voltage Regulators 1A, 3.3V	U2	FP-AMS1117-IPC_C	CMP-209535-000001-1	1
USBLC6-2SC6	ESD Protection for USB 2.0 High Speed	U3	SOT23-6L_STM	USBLC6-2SC6	1
ABM3B-8.000MHZ-B2-T	Crystal 8MHz ±20ppm 18pF SMD-4 5mm x 3.2mm	Y1	FP-ABM3B-MFG	CMP-1688-00002-2	1
ABS07-120-32.768KHZ-T	CRYSTAL 32.7680KHZ 6PF SMD	Y2	FP-ABS07-120-MFG	CMP-2000-07587-2	1